

DAVID LIU

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| Claremont, CA, USA | LinkedIn

PROFESSIONAL SUMMARY

Mechanical Engineering student at Harvey Mudd College with experience in CAD, FEA, CFD, manufacturing, and autonomous systems. Interested in aerospace, defense, robotics, and advanced vehicle design.

EDUCATION

Harvey Mudd College

Bachelor of Science (BS), Mechanical Engineering

August 2024 - May 2028

GPA: 3.74

SKILLS

CAD & Design: SolidWorks, Engineering Drawings, GD&T, Materials Selection, Circuit Design

Simulation & Analysis: FEA, CFD, MATLAB, Data Analysis, COMSOL

Fabrication: Machining, 3D Printing, Mechanical Assembly, Prototyping

Programming: Python, MATLAB, C++, Java, LabVIEW, Arduino IDE

Languages: Spanish, Chinese, English

PROFESSIONAL EXPERIENCE

MRP Solutions

Electro-Mechanical Engineering Intern

June 2026 - Present

Plattsburgh, NY, USA

- Designed 10+ mechanical assemblies, machine guards, and access-control systems in SolidWorks for automated manufacturing equipment, producing fabrication-ready models and engineering drawings.
- Engineered retrofit locking systems and custom sheet-metal components for production machinery, translating engineering requirements into manufacturable hardware solutions.
- Performed troubleshooting and root-cause analysis on integrated electro-mechanical systems, diagnosing mechanical, electrical, and process-related failures.
- Conducted equipment testing, calibration, and validation activities to identify performance issues, support engineering modifications, and improve operational reliability.
- Collaborated with mechanical, electrical, controls, and operations teams to implement hardware improvements and support system integration.

Harvey Mudd College

Lab TA/Tutor/Grader

August 2025 - Present

Claremont, CA, USA

- Assisted instruction in chemistry laboratories and supported experimental troubleshooting and safety compliance.
- Tutored students in Python, MATLAB, calculus, and chemistry, strengthening technical communication and problem-solving skills.

ENGINEERING EXPERIENCE

Electro-Mechanical Engineering Intern

May 2026 - Present

- Designed 10+ mechanical assemblies, machine guards, and access-control systems in SolidWorks for automated manufacturing equipment, producing fabrication-ready models and engineering drawings.
- Performed troubleshooting and root-cause analysis on integrated electro-mechanical systems, diagnosing mechanical, electrical, and process-related failures.
- Conducted equipment testing, calibration, and validation activities to identify performance issues, support engineering modifications, and improve operational reliability.
- Collaborated with mechanical, electrical, controls, and operations teams to implement hardware improvements and support system integration.

Autonomous Underwater Robot with Multi-Sensor Environmental Monitoring

January 2026 - May 2026

- Designed and integrated pressure, pH, turbidity, and temperature sensing systems for an autonomous underwater vehicle (AUV) enabling environmental data collection to depths of 2 meters.
- Developed and calibrated analog signal-conditioning and closed-loop depth control systems on a Teensy 4.0, using MATLAB regression models and Arduino-based feedback control for stable autonomous operation.
- Conducted ocean field testing, validating sensor performance and analyzing water-quality trends across turbidity, pH, temperature, and depth data.

Harvey Mudd RoboSub Team

August 2024 - Present

- Designed and prototyped a torpedo release mechanism for an autonomous underwater vehicle, focusing on reliability in submerged conditions

- Performed FEA in COMSOL to evaluate stress concentrations and deformation under operational loading conditions, informing redesign decisions that improved structural reliability.
- Redesigned PCB layout to improve signal routing reliability and reduce electrical noise in underwater environments

Mudd Automotive Club: Hybrids

August 2024 - Present

- Contributed to the design and manufacturing of a student-built hybrid vehicle, supporting powertrain integration, braking systems, and steering architecture
- Designed and fabricated structural frame and front-end assemblies, including brake mounts and steering components
- Developed detailed SolidWorks models and assemblies, translating designs into manufacturable parts through machining and iterative prototyping
- Participated in hands-on assembly and system testing

INVITE Institute REU, UIUC

- Developed LLM-based tools to analyze structured student performance datasets
- Designed synthetic datasets to evaluate model robustness and retrieval accuracy
- Built Python pipelines for data cleaning, structuring, and statistical evaluation
- Conducted quantitative analysis using performance metrics and error evaluation

DOFPro Project, Harvey Mudd College

- Created interactive technical materials for Chemical & Thermal Processes using Reveal.js and LaTeX
- Translated complex engineering concepts into structured visual and computational models
- Published materials to faculty platform: dofpro.org

Independent BioChem Research

- Investigated resiniferatoxin applications for osteoarthritis pain treatment
- Conducted literature analysis and synthesized findings into structured technical documentation
- [Link to project](#)

Computational and Experimental Aerodynamic Analysis of Nose Cone Geometries

January 2026 - May 2026

- Designed and executed a computational and experimental study comparing aerodynamic performance across multiple nose cone geometries.
- Developed CFD models in COMSOL Multiphysics and validated predictions against wind tunnel testing, evaluating flow behavior, pressure distributions, drag characteristics, and model accuracy.
- Applied fluid mechanics principles to analyze Reynolds number effects and identify geometry optimization strategies for drag reduction and improved aerodynamic performance.

RESEARCH

INVITE Institute REU, UIUC

May 2025 - August 2025

- Developed LLM-based tools to analyze structured student performance datasets
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